MIL-T-46333A(MU)
10 September 1970
SUPERSEDING
MIL-T-46333(Ord)
9 March 1962

MILITARY SPECIFICATION

TELESCOPE, ELBOW: M101

1. SCOPE

1.1 This specification covers one type of elbow telescope known as Telescope, Elbow: M101, hereinafter referred to as "telescope". This telescope is used for laying the 105mm Self-Propelled Howitzer, M52 in direct fire.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments: General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

C8213267	Center Ground Telescope
D7695482	Fixture Measuring Elevation and Azimuth Error

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D7560085	Fixture Vibration
D7695509	Fixture to Check Image Tilt
F7659479	Final Inspection Fixture
F7695945	Telescope, Elbow M101
F8565556	Fixture Pressure Tester
F8614973	Adapter, Vibration

PACKAGING DATA SHEET

7695945	Packaging of Telescope, Elbow M101

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The telescope shall be manufactured in accordance with Drawing F7695945 and drawings pertaining thereto.

3.2 General specifications.- The contractor shall be responsible for adherence to and compliance with the requirements of the general specifications specified below. Should any conflict exist between these general specifications and this specification, this specification shall take precedence.

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Electrical and electronic assemblies
e. Part identification and marking
f. Workmanship

3.2.2 MIL-O-13830.

a. Cleanliness
b. Illumination of reticle
c. Resolution

3.3 Temperatures.

3.3.1 Storage temperature.- The telescope shall be exposed and thermally stabilized at ambient temperatures of minus 80 degrees and plus 160 degrees Fahrenheit (F). Upon return to standard ambient temperature +60° to 90°F) the telescope shall show no indication of physical failure and shall meet the performance requirements of this specification.

3.3.2 Operating temperature.- The telescope shall not exceed the maximum torque requirements of 3.8 when exposed and thermally stabilized at ambient temperatures of -40°F and +150°F.

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3.4 Vibration.- The telescope shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes ± 15 seconds. Subsequent to vibration, the telescope shall show no evidence of loose or damaged parts and shall meet the requirements of 3.5 to 3.9 inclusive.

3.5 Cleanliness.

3.5.1 Dirt.- Inspection for dirt and foreign particles (classed as dirt) shall be made from the eye end only. The maximum size of any particle appearing on the reticle shall be a reticle line width. The maximum number of dirt particles appearing within a central field of view of 50 mils diameter shall be 3. The maximum number of dirt particles appearing outside this field of view shall be 3. The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution.

3.5.2 Other defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations when viewing through the objective and eye end of the instrument.

3.6 Sealing.- The telescope, when subjected to an internal pressure of 5 pounds per square inch gage pressure (psig) of dry nitrogen shall show no evidence of leakage in excess of 0.20 psig over a period of 2 hours. The interior of the accepted telescope shall contain dry nitrogen having a dew point of at least as low as -25°F at a pressure of between 0.2 to 0.5 psig.

3.7 Performance.

3.7.1 Collimation.- With the axis of the trunnions horizontal, the telescope positioned in accordance with Figure I, the cant level bubble centered, and the boresight knobs set to zero index, the boresight cross shall be 40 mils ± 1.0 mil above the geometric axis, and shall coincide within 1.0 mil of a plumb line passing through the geometric axis.

3.7.2 Circular field of view.- With the deflection and elevation boresight knobs at zero index, the horizontal and vertical fields of view shall not differ by more than 2.5 degrees.

3.7.3 Parallelism of reticle and image.- With the telescope positioned as in 3.7.1, the vertical reticle line shall be parallel to the image of a vertical target line within 40 minutes of arc.

3.7.4 Image tilt.- The image of a plumb line shall be vertical within 2 degrees.

3.7.5 Elevation accuracy.- The line of sight through the boresight cross shall remain in coincidence with its target aiming point within ±0.5 mil in elevation and deflection (azimuth) when elevating the objective assembly from -10° to +65°.

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3.7.6 Boresight knob travel.

3.7.6.1 Deflection.- Rotation of the deflection boresight knob shall move a target image a minimum of ± 10 mils in a horizontal plane from the position established in 3.7.1.

3.7.6.2 Elevation.- Rotation of the elevation boresight knob shall move a target image a minimum of ±10 mils in a vertical plane from the position established in 3.7.1.

3.7.7 Cant correction.

3.7.7.1 Angular movement.- With the trunnions canted 15 degrees to the right and 15 degrees to the left, rotation of the cant correction knob shall reposition the vertical line to be parallel to a vertical target line within ±2.0 degrees when level vial is centered.

3.7.7.2 Boresight movement.- With the trunnions canted 15 degrees to the right and 15 degrees to the left, rotation of the cant correction knob shall not cause the boresight cross to shift by more than 0.25 mils when the level vial is centered.

3.7.8 Image shift.- Actuation of the boresight locking levers to the locked position shall not change position of the target image in excess of 0.1 mil.

3.7.9 Eyepiece focus.- Eyepiece focus shall be fixed between the limits of -0.75 and -1.0 diopter at which setting the reticle shall appear in sharp focus.

3.7.10 Resolution.- The resolving power on the optical axis shall be 12 seconds of arc or less.

3.7.11 Parallax.- Parallax at the center of the field shall not exceed 0.15 mil when viewing a target at 300 yards ± 10 yards.

3.7.12 Reticle accuracy.- The angular value in mils between the zero and maximum range line shall be as indicated on the reticle drawing within ±1.0 mil.

3.8 Operability.- The running torque required to rotate the cant correction knob and the boresight knobs, with locks fully released shall be 4 to 12 inch pounds when checked at standard ambient temperature. The values obtained at extreme temperatures as specified in 3.3.2 shall not exceed 25 inch pounds. With the boresight knobs locked, the knob shall not move when a torque of 20 pound inches is applied at standard ambient temperature.

3.9 Illumination.- With the reticle and level vial lamps energized with 3 volts, the reticle markings and level vial shall appear clearly defined when observed in a darkened area.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any

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commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) telescopes plus three (3) each of all items covered by SQAP when SQAPs are provided. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests,

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

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4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Table I herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table I shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: 100% Inspection		
1. Sealing	3.6	4.6.1.2
MAJOR: 1.0%		
101. Collimation pre-vibration	3.7.1	4.6.2.1.1
102. Vibration	3.4	4.6.2.1.2
103. Collimation post-vibration	3.7.1	4.6.2.1.4
104. Cleanliness	3.5	4.6.2.1.3
105. Circular field of view	3.7.2	4.6.2.2
106. Boresight movement	3.7.7.2	4.6.2.3
107. Parallelism of image and reticle	3.7.3	4.6.2.4
108. Image tilt	3.7.4	4.6.2.8
109. Elevation accuracy	3.7.5	4.6.2.14
110. Boresight travel (deflection)	3.7.6.1	4.6.2.5
111. Boresight travel (elevation)	3.7.6.2	4.6.2.6
112. Angular movement	3.7.7.1	4.6.2.11
113. Image shift	3.7.8	4.6.2.7
114. Eyepiece focus	3.7.9	4.6.2.9
115. Resolution	3.7.10	4.6.2.13
116. Parallax	3.7.11	4.6.2.12
117. Reticle accuracy	3.7.12	4.6.2.10
118. Operability	3.8	4.6.2.15
119. Illumination	3.9	4.6.2.16

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling for environmental testing.

4.4.1 Samples shall be selected from lots at random using inspection level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements and tests outlined in Table I prior to being subjected to the following examination and test. The requirements outlined in 3.1 and 3.2 shall be accepted on contractor's certification.

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TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Temperature (operating)	3.3.2	4.6.1.1
302. Temperature (storage)	3.3.1	4.6.1.1
303. Operability (operating temperature)	3.8	4.6.2.15
304. Torque (operating temperature)	3.8	4.6.2.15

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6 and 4.7.

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NOMENCLATURE	DESCRIPTION
1. Hot and cold chamber	Standard type conforming to the accuracies outlined in test facilities specified under test facilities of MIL-F-13926
2. Power supply	0 to 30 vdc output within 10% accuracy
3. Torque wrench and scale	Standard shop type accurate to within 10% of drawing tolerance

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Temperature test.- The telescope shall be subjected to test Procedure I of MIL-F-13926; however, the temperatures specified in 3.3.1 and 3.3.2 shall apply. When the telescope has been stabilized at the temperature specified, the result of the specified test shall conform to the limits outlined in 3.3.1 and 3.3.2 using test procedures outlined herein.

4.6.1.2 Sealing.- This test shall be performed using testing equipment the accuracy of which shall conform to the accuracy depicted on Drawing F8565556. When using pressure test fixture, the fixture shall be set up as outlined in set-up instructions shown on Drawing F8565556, Sheet 1. The gas used shall be dry nitrogen having a dew point of -25°F. After connecting the test fixture to the telescope, adjust pressure for time specified in 3.6. Pressure gage shall remain constant within the allowable leakage specified in 3.6. At the completion of this test, connect a dew point tester to the exit post and purge the optical assembly until -25°F is recorded and seal the instrument. After completion of this sealing test, measure the internal pressure using standard shop type test equipment and inspection procedures for conformance to internal pressure specified in 3.6.

4.6.2 Performance testing.

4.6.2.1 Orientation.- The telescope under test shall be properly oriented when positioned on Fixture F7695479 in accordance with the set-up instruction specified thereon.

4.6.2.1.1 Collimation.- With the telescope oriented as outlined in 3.7.1, sight through the eyepiece and observe where the boresight cross falls upon the target point representing the telescope's optical axis with respect to the two mil circle. Record this position after having locked all knob locks.

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4.6.2.1.2 Vibration.- This test shall be performed using a testing device, the accuracy of which shall be equal to or exceed the accuracy depicted on Fixture F7560085 and Adapter F8614973. The test shall be performed as outlined in Procedure V of MIL-F-13926; however, the duration, frequency and amplitude shall be as specified in 3.4. At the conclusion of the test, the telescope shall be subjected to a visual and tactile examination and shall be subjected to the following performance test to determine conformance with 3.4.

4.6.2.1.3 Cleanliness.- The telescope shall be tested for cleanliness by using the technique outlined in MIL-O-13830. At the conclusion of this test, dirt particles shall not exceed the limits outlined in 3.5.

4.6.2.1.4 Post vibration.- Immediately after completing the vibration test, reassemble the telescope to the fixture and without disturbing any previous settings, observe where the telescope's boresight cross falls upon the target point outlined in 4.6.2.1.1. This observed point and the recorded position shall coincide within the limits specified in 3.7.1.

4.6.2.2 Circular field of view.- With the telescope positioned as outlined in 4.6.2.1, sight through the telescope and observe the total field of view. Ellipticity (if any) in the total field of view shall not exceed the limits outlined in 3.7.2. Use target lines (arc lines) when determining the limits.

4.6.2.3 Boresight movement.- Position the telescope in the test fixture using one of the appropriate "V" blocks (either Item 26 or 27) shown on Drawing 7695479 under the telescope's trunnions to provide either the right or left cant position. Using the boresight knobs, position the intersecting point of the boresight cross close to but not touching the corresponding target point. Sight through the telescope and rotate the cant knob in the direction to remove the cant. Observe displacement, if any, between the points in coincidence. They shall not move more than the limits outlined in 3.7.7.2. Use the thickness of the reticle line (2 1/2 times thickness) to determine extent of acceptable movement. Repeat this procedure using the other "V" block and with the telescope canted in the opposite direction.

4.6.2.4 Parallalism of reticle and image.- With the telescope positioned as outlined in 4.6.2.1, sight through eyepiece and observe parallelism between center vertical range lines and the plumb line. By means of the boresight knobs, position one end of the center vertical range line close to but not touching the plumb line and observe parallelism at the opposite end of this range line with respect to plumb line. Both must be parallel within the limits of target pip lines provided for this purpose to assurance conformance with 3.7.3.

4.6.2.5 Boresight (deflection) knob travel.- With the telescope positioned as outlined in 4.6.2.1, observe coincidence between the boresight's intersecting point and the point representing the target point of collimation. Record this position. Observe where this intersecting point falls with respect to the 20 mil diameter tolerance circle after the deflection knob has been rotated 10 mils as measured by the deflection scale. This intersecting point must travel 10 mils from the recorded position. Use the 20 mil diameter

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target circle when determining conformance with 3.7.6.1. Repeat this observation after having rotated the deflection knob in the opposite direction from the point of collimation (4.6.2.1).

4.6.2.6 Boresight (elevation) knob travel.- With the telescope positioned as outlined in 4.6.2.1, repeat the procedure outlined in 4.6.2.5 except the elevation knob will be used. The intersecting point of the boresight cross shall travel 10 mils from the recorded position, use the 20 mil target circle to assure conformance with 3.7.6.2.

4.6.2.7 Image shift.- With the telescope positioned as outlined in 4.6.2.1, sight through the telescope and by means of the boresight knobs, position the boresight cross vertical line close to but not touching the plumb line and position the boresight cross horizontal line close to but not touching the horizontal target line. Record this position. Lock boresight knobs; don't disturb boresight knobs. Sight through the eyepiece and observe where boresight cross is with respect to the plumb line and target horizontal line. Movement, if any, shall not exceed the limits of 3.7.8. Use thickness of reticle line when calculating reticle shift.

4.6.2.8 Image tilt.- With the telescope positioned as outlined in 4.6.2.1, align an auxiliary scope D7695509 that has been positioned as outlined in the set-up and operating instructions. Position the intersecting point of the auxiliary telescope's reticle on the intersecting point of the top horizontal target line and plumb line while simultaneously maintaining auxiliary telescope's reticle plumb. Observe where the plumb line falls with respect to tolerance lines of the auxiliary telescope's reticle. The plumb line must not exceed the limits of the reticle angular lines to assure conformance with 3.7.4.

4.6.2.9 Eyepiece focus.- With the telescope positioned as outlined in 4.6.2.1, place a while background in front of the objective assembly and remove the diaphragm from the telescope objective assembly. Set a dioptometer to zero diopters and at the parallax distance,and also focus the eyepiece to the individual inspector's eye. Align the dioptometer's optical axis to the optical axis of the telescope under test. Adjust the dioptometer to obtain the clearest sharpest image of the telescope's reticle. When this adjustment has been obtained, record the diopter value shown on the dioptometer. This reading must be within the limits specified in 3.7.9..

4.6.2.10 Reticle accuracy.- With the fixture and telescope positioned as outlined in 4.6.2.1, sight through the eyepiece assembly and observe when the specified range line noted in 3.7.12 falls upon the comparable target range line, when the boresight cross is in coincidence with the horizontal target line and plumb line. They must coincide within the limits of the target tolerance lines to assure conformance with 3.7.12.

4.6.2.11 Angular movement.- Position the telescope in the test fixture using one of the appropriate "V" blocks (either Item 26 or 27) shown on Drawing 7695479 under the telescope's trunnions to provide either the right or left cant position. Rotate the cant knob in the opposite direction and level the cant level vial. Observe parallelism between the vertical reticle line and the approriate target line. They must be parallel within the target's angular tolerance line to assure conformance with 3.7.7.1. Repeat this procedure with the telescope canted in the opposite direction.

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4.6.2.12 Parallax.- Remove the telescope from the fixture, adjust and assemble the Parallax Tube 7695482 as outlined in the set-up and operating instructions. Sight through the eyepiece end and by means of the boresight knobs, set the center vertical deflection line and the 1200 yard range line close to but not touching the comparable reticle lines of the parallax tube. Move the head side to side and up and down. There shall be no movement at the optical axis of the telescope in excess of the limit specified in 3.7.11. Use the thickness of the telescope reticle line when estimating reticle movement. Movement, if any, shall not exceed the telescope's reticle line by 1 1/2 times the thickness.

4.6.2.13 Resolution.- Remove the telescope from the fixture and remove the diaphragm. Support the telescope using appropriate means and direct the line of sight toward an appropriate definition target. The test bar pattern and the target distance together with the test procedure shall be as outlined in MIL-O-13830 to assure conformance with 3.7.10.

4.6.2.14 Elevation accuracy.- With the telescope removed from the fixture, remove diaphragm and secure Fixture 7695482 to the telescope as outlined in the set-up and operating instructions. Sight through the eyepiece and position the intersecting point of the boresight cross central with respect to the limit line of the fixture reticle. Use boresight knobs to obtain this position. While sighting through the eyepiece, raise and lower fixture through the angular excursion outlined in 3.7.5 and observe movement of the boresight's intersecting point with respect to the set position. Movement, if any, shall not exceed the reticle tolerance lines to assure conformance with 3.7.5.

4.6.2.15 Operability.- Using tactile inspection techniques, determine all knobs rotate within the confines of normal instrument shop practices. Actuate the boresight locking levers. Using standard inspection procedures together with appropriate torque wrench and adapters, determine running torque values at the temperature specified in 3.8.

4.6.2.16 Illumination.- With 3 volts applied to the telescope, observe the illumination of the reticle with the objective end covered. Also, observe illumination of the level vial. Observation of level vial illumination may require the use of a darker area. All graduation must appear as specified in 3.9.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The telescope shall be packaged, packed and marked in accordance with Packaging Data Sheet 7695945. The level of protection shall be as specified in the procurement documents.

6. NOTES

6.1 Intended use.- The Elbow Telescope: M101 formerly designated T150E1 is primarily intended to lay the 105mm armament of the Self-Propelled Howitzer, M52 in direct fire. A unity power periscope known as Periscope:

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M23 (formerly T38) locks on the forward assembly of the telescope mount and over the objective end of the telescope thus completing the objective section of the direct fire control system.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing
c. Applicable stock number
d. Applicable packaging data sheet number (see 5.1)
e. Provisions for First Article testing

Custodian:

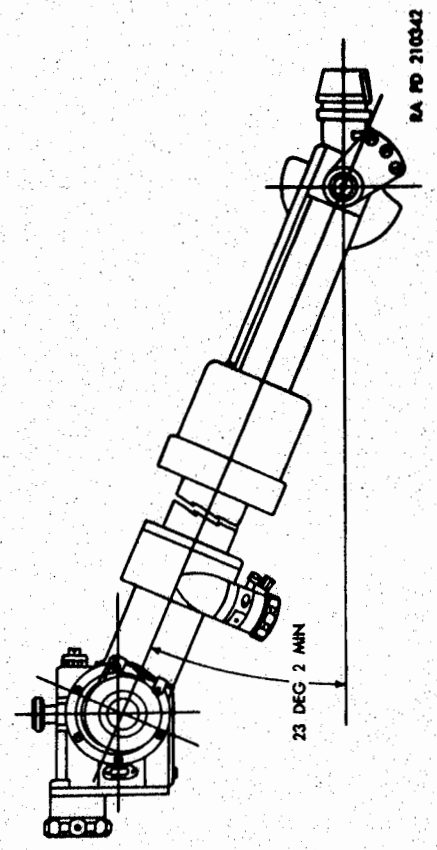
Army - MU

Preparing activity:

Army - MU

Project No. 1240-A257

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TELESCOPE, ELBOW: M10I

FIGURE 1.

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